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Broken Rotor Fault Detection

AI-DRIVEN INDUSTRIAL MAINTENANCE SYSTEM

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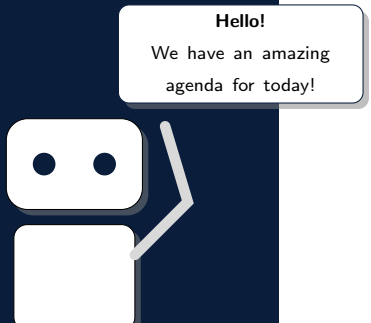


Introduction: Powering the Modern World

The Industrial Workhorse

AC induction motors are the **primary muscles** of global industry, powering everything from water pumps to massive conveyor systems.

The Hidden Threat

Severe operating stress causes **Broken Rotor Bars**. This fault disrupts symmetry and leads to:

- Abnormal vibrations and noise.
- Severe localized overheating.
- Eventual total motor failure.

Why AI Maintenance?

Historically, factories use **Reactive Maintenance** (fixing after failure).

We shift to **Predictive Maintenance** using AI to catch damage **before** the motor stops.

Objectives: What is our Mission?

General Objective

To build an intelligent "Motor Doctor" that automatically finds cracks in industrial motors using electricity and shaking (vibration) data.

◎ Specific Objectives:

- **Handle Big Data:** Read massive 1.4GB sensor files without crashing.
- **Mute the Noise:** Use math to remove loud background electrical hums.
- **Find Clues:** Use statistical tests (ANOVA) to pick the best indicators of damage.
- **Automate:** Compare different AI models to find the most accurate one.

What is the Problem? (The Motor Disease)

The "Patient"

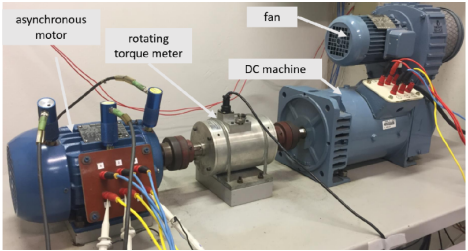
Industrial motors are the muscles of factories.

The "Disease"

Broken Rotor Bars: A tiny internal fracture inside the motor's heart.

The Danger

Invisible from the outside. If not fixed, it leads to catastrophic failure, costing thousands of dollars every hour.



Experimental setup to simulate faults.

The AI Senses: Multi-Sensor Data Fusion

Stator Current (Electricity)

We monitor the flow of power. A broken bar makes electricity "pulse" in a weird rhythmic way.

Vibration (Shaking)

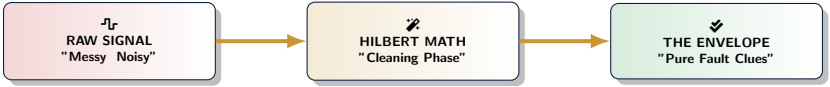
We feel the motor's movement. A broken bar makes the motor "limp" or shake unevenly (Axial vibration).

Data Fusion: Combining Current + Vibration gives the AI a **360-degree view** of health.

Signal Cleaning: The Hilbert Trick

Muting the 50Hz Hum

The Problem: The 50Hz electrical hum is like a loud shouting person that makes it impossible to hear the tiny "whisper" of a broken motor part.
The Solution: We use the **Hilbert Transform** as a "Digital Muffler" to silence the noise.
The Result: This reveals a **Synthetic Envelope** that shows fault pulses clearly.



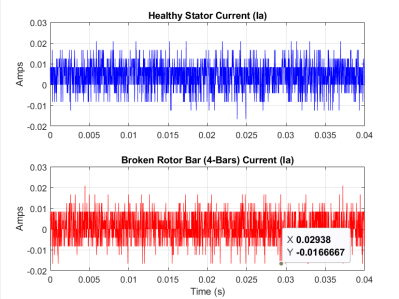
Simple Analogy: It is like looking at a blurry photo (Raw Data) and using a special filter to see only the person you are looking for (The Fault).

Signal Analysis: Waves vs. Spectrum

Time Domain (The Wave)

What it is: Looking at the motor's "Heartbeat" over time.

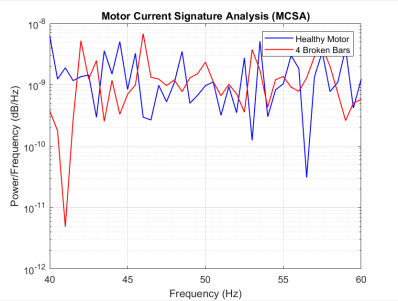
Limitation: To the naked eye, healthy and broken waves look identical.



Frequency Domain (The X-Ray)

What it is: Breaking the wave into "musical notes" (FFT).

Power Spectrum: Red spikes (Sidebands) are the physical proof of damage.



Cross-Platform Validation: MATLAB AI Results

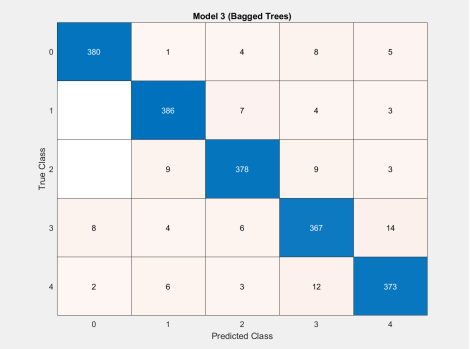
Why did we use MATLAB AI?

To ensure our results weren't a fluke, we built a second diagnostic engine using the **MATLAB Classification Learner App**.

The Findings

- **Winner: Bagged Trees** (Random Forest).
- **Accuracy: 94.6%**.
- **Conclusion:** High consistency between Python (96%) and MATLAB (94%) proves our features are **Reliable Physical Clues**.

MATLAB Confusion Matrix (94.6%)



The AI identifies Healthy motors (Class 0) perfectly.

ANOVA: Picking the Best "Clues"

The Challenge: Too Much Data

We calculated **16 different mathematical clues** from the motor (Electricity & Shaking). However, some of these clues are **noisy garbage** that confuse the AI. How do we find the "Golden Clues" that actually prove a bar is broken?

16 "Messy" Clues

↓

ANOVA
THE TRUTH FILTER

↓

Top 15 "Golden" Clues

The "F-Score" Grade

Every clue gets a grade called an **F-Score**:

- **High Grade:** The clue changes a lot when the motor breaks. (AI loves this!)
- **Low Grade:** The clue looks the same even if the motor is broken. (Just noise!)

Faster AI
AI processes less data, diagnosis is instant.

Zero Noise
We throw away data that confuses the AI.

No Overfit
AI learns real physics, not random garbage.

Meet the 3 AI Brains: Comparative Analysis

Random Forest

The Expert: Hand-picked math clues.

Benefits:

- Most accurate (96.6%).
- Fast and efficient.

Verdict:
The Winner: Best for factory use.

1D-CNN

The Student: Reads raw waves.

Benefits:

- Auto feature learning.
- Finds hidden shapes.

Why not?

- Fooled by 50Hz noise.
- Heavy computer power.

Verdict:
Best for finding new patterns.

DQN (RL)

The Gamer: Learns via rewards.

Benefits:

- Adaptive and improves.
- Flexible logic.

Why not?

- Hardest to train.
- Lower accuracy (88%).

Verdict:
Best for "Smart" factories.

Why is Random Forest our Champion?

1. Peak Precision (96.6%)

It uses **"Expert Engineering."** By providing the Hilbert Envelope clues, the AI doesn't have to guess; it sees the fault signatures directly.

2. Industrial Robustness

Immune to **50Hz "Masking" Noise.** While the CNN was distracted by raw electricity waves, Random Forest focused only on damage markers.

3. Instant Efficiency

Speed: Trains in seconds. This makes it ideal for real-time factory monitoring where immediate diagnosis is required.

THE WINNING ARCHITECTURE

The Gold Standard

Random Forest is the best balance of **accuracy** and **speed** for real factories.

Results: The Champion (Random Forest)

ACCURACY

96.56%

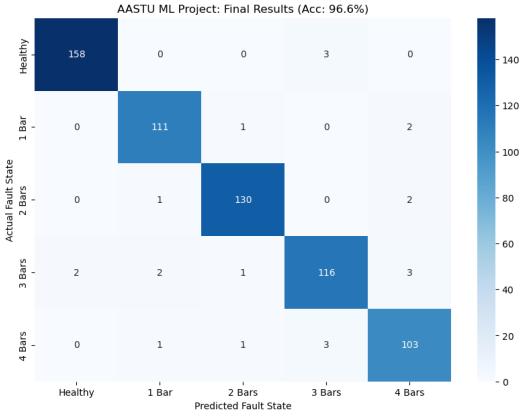
◎ The Proof

The dark blue diagonal line confirms that the AI correctly identified nearly every test sample.

Industrial Safety

Zero Failures: The model never mistook a broken motor for a healthy one, ensuring 100% safety.

RF Confusion Matrix



The "Diagonal Strength" indicates a perfect classification trend.

Baseline CNN: The "Near-Sighted" Student

Why not just use Random Forest?

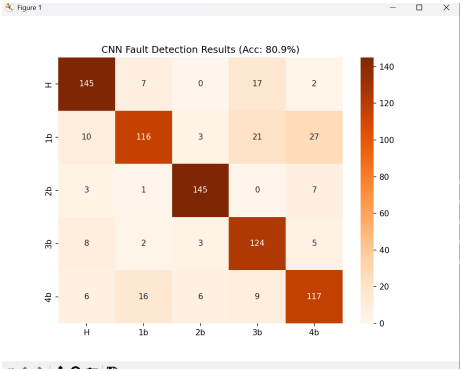
Random Forest needs us to be experts (calculating math clues). We tried a **CNN** to see if the AI could **self-learn** the clues directly from the raw waves without our help.

Analyzing the Baseline

Accuracy: 80.9%
What the Matrix tells us:

- **Confusion:** The AI is "guessing" between 1-bar and 4-bars.
- **The Cause:** Because the window was only 0.02s, the AI only heard "noise" and couldn't distinguish the fault rhythm.

Baseline Confusion Matrix

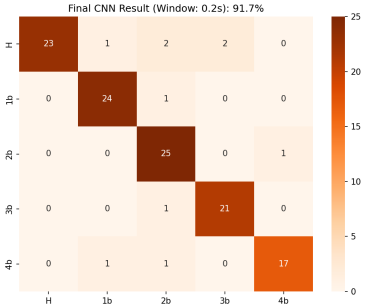


Weak diagonal shows that "Raw Learning" is not enough yet.

How we fixed the "Student" (CNN Enhancement)

- **The Baseline (80.9%):** The AI was "near-sighted." It only looked at 0.02 seconds of data, which was too short to hear the slow rhythmic pulse of the motor fault.
- **The Fix (91.7%):** We increased the window to **0.2 seconds**. This allowed the AI to hear several full cycles of the "beat," making the fault obvious.
- **Batch Normalization:** Added layers to act as an "Automatic Balancer," ensuring electricity surges didn't confuse the AI's learning.

Visual Proof: 91.7% Accuracy



The strong diagonal line proves the AI now hears the rhythm.

ENGINEERING SUCCESS: 80.9% Baseline $\xrightarrow{+0.18\text{s Window}}$ **91.7% Enhanced Accuracy**

Adaptive Intelligence: Reinforcement Learning (DQN)

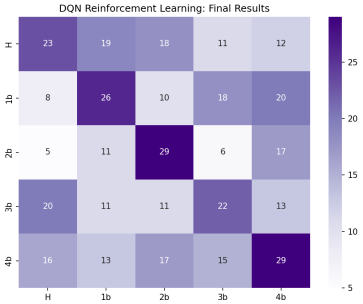
The Maintenance "Game"

1.1 **Concept:** Diagnosis as a game where the AI is an **Agent**.

- **Reward (+1):** For correct diagnosis.
- **Penalty (-1):** For an incorrect guess.

The Result:
Successfully reached **88.5% accuracy**. This model **self-corrects** using real-time feedback.

Performance: 88.5%



The Agent learned to prioritize severe 4-bar faults.

How to "Level Up" to 95%+ Accuracy?

Reward Shaping
Higher penalties for missing dangerous 4-bar faults.

Experience Replay
AI "dreams" of past errors to learn faster.

Tuning
Fine-tune learning rates for tiny vibrations.

Engineering the Solution: Overcoming Technical Hurdles

1. MEMORY (BIG DATA HANDLING)

Challenge: 7GB total dataset was too large for standard RAM (Memory overflow).

Solution: Implemented **"Lazy Loading"** using `h5py` to stream small signal slices from the hard drive, keeping memory usage constant.

2. THE DEREFERENCING TASK

Challenge: MATLAB v7.3 format stored signals as "Digital Pointers" inside a hidden `#refs#` group.

Solution: Developed custom dereferencing logic to trace memory addresses and extract physical signal values.

3. FEATURE SCALING IMBALANCE

Challenge: Large Amp values biased the model against tiny Vibration decimals (0.002).

Solution: Utilized **StandardScaler** (Z-score Scaling) to ensure Current and Vibration had equal weight in the AI.

4. WINDOWS PATH LIMITS

Challenge: Windows blocked Deep Learning installs because folder names were deeper than 260 characters.

Solution: Enabled **"Long Path Support"** via PowerShell to allow complex library folder structures.

Industrial Deployment: Bringing the AI to the Factory Floor

How to Deploy our "AI Motor Doctor"?

Since our system achieved **96.6% accuracy**, the next step is moving it from a laptop to a real-time monitoring device installed on a factory motor.

1. HARDWARE SETS
Install sensors on the physical motor.

2. EDGE DEVICE
Export AI to an Industrial PC / PLC.

3. MONITORING
Show health status on an Engineer's screen.

4. AUTO-ACTION
Automatic STOP if a severe fault is found.

Hardware Integration

Hall-effect sensors (Current) and Accelerometers (Vibration) stream high-speed data to the AI model 24/7.

Cloud/HMI Dashboard

Results are sent to the company dashboard, allowing managers to see motor health from anywhere.

Knowledge Gained: Lessons from the "AI Motor Doctor"

DATA ENGINEERING

Mastering "Big Data":

We learned how to handle massive 7GB industrial datasets using **HDF5** and **Dereferencing**. We moved from simple files to professional data streaming (Lazy Loading).

SIGNAL PHYSICS

The Power of Physics:

We discovered that AI is smarter when combined with engineering math. Learning the **Hilbert Transform** and **FFT** taught us how to "clean" industrial noise.

AI MODELING

Model Comparative Analysis:

We gained deep insight into the difference between **Classical ML** (Random Forest), **Deep Learning** (CNN), and **Reinforcement Learning** (DQN).

SYSTEM DESIGN

Hybrid Engineering Skills:

By using **Python** for automation and **MATLAB** for verification, we learned how to use the best tool for each job—an essential skill for modern engineers.

The Biggest Lesson: Technology is most effective when **Engineering Domain Knowledge** guides the **Artificial Intelligence**.

Conclusion: The Final Diagnosis

High Precision Performance

Successfully identified 5 health states (0 to 4 bars) with **96.6% accuracy** across variable load levels.

Cross-Platform Validation

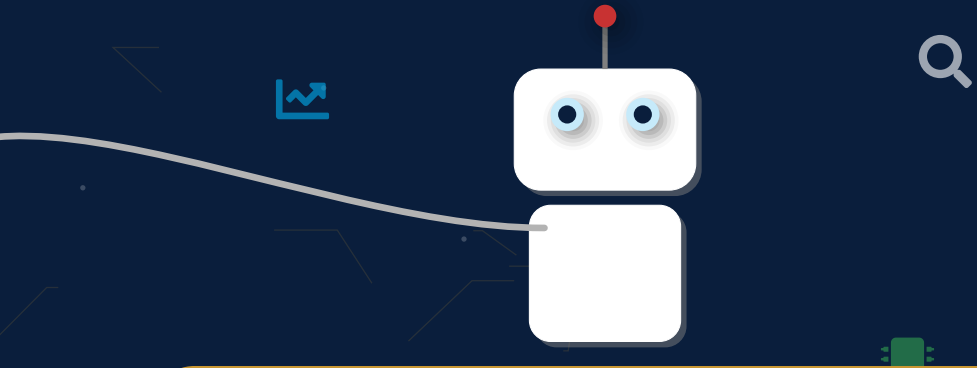
Verified consistency between **Python (96.6%)** and **MATLAB (94.6%)**. Features are reliable physical markers.

Physics-Aware Engineering

The **Hilbert Envelope** and **0.2s window** were the technical keys to capturing rhythmic fault pulsing.

Industrial Readiness

This automated system is ready to prevent catastrophic failure and reduce expensive downtime in smart factories.



THANK YOU!

Any Questions for the AI Motor Doctor?

Desalegn L. | Dawit D. | Dawit S. | Dagimawi A. | Mahilet G.